Development of an 8-channel EEG acquisition Brain Computer Interface System

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Introduction

The primary objective of this project is to develop a circuit capable of acquiring 8 separate EEG channels through electrode connections. The system design is divided into two distinct iterations to explore different architectural approaches. The first version (M1) utilizes a discrete component approach, where each channel

consists of independent stages for instrumentation amplification, filtering (high pass, low pass, and notch), and a final amplification stage. The second version (M2) integrates these functions using the ADS1299, a component renowned for biosignal acquisition applications.
In both iterations, the signal undergoes conditioning and amplification before being directed to an Analog-to-Digital Converter (ADC). This digital signal is then forwarded to an interface for visualization. Power management differs between the two; the discrete circuit (M1) is powered by 9V batteries, whereas the integrated version (M2) operates via USB power. To maintain signal integrity and safety, the first board implements a Right Leg Driver (RLD) circuit, while the second utilizes a Bias configuration consistent with the ADS1299 datasheet specifications.

M1 Analog Front End

Protection

Given that EEG systems interface directly with human subjects, incorporating circuit protection is critical. To address this, Texas Instruments' TPD4E1B06DCKR, a 4-channel bi-directional Transient Voltage Suppressor (TVS), was implemented. Three of these ICs were utilized to provide electrostatic discharge protection for all 8 signal channels, as well as the reference and bias lines, isolating the subject from the circuit components.

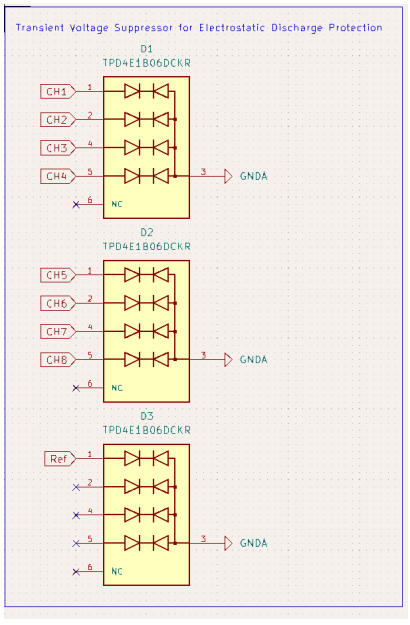


Fig 1. TVS Array for M1 in KiCAD

RF Filters

To further protect the sensitive EEG signals from high-frequency interference (MHz-GHz range) typical of broadcast and wireless communications, Radio Frequency (RF) filters were introduced . specifically, EMI LCL filters were placed in the pre-amplification stage for all 8 channels, the reference, and the bias lines.

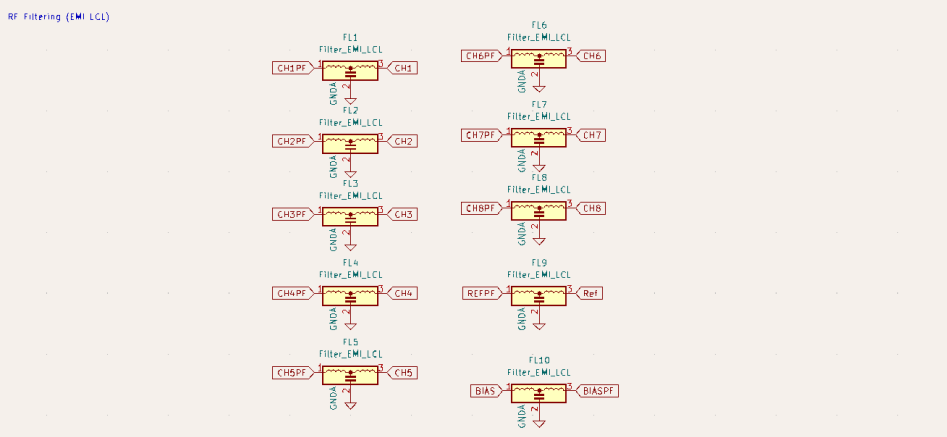


Fig 2. EMI LCL in KiCAD

Instrumentation Amplifier

Following the protection stages, the signal enters the initial acquisition phase. The system utilizes Analog Devices' AD8422, a rail-to-rail precision instrumentation amplifier known for high performance and low power consumption. This component provides the initial gain before filtering occurs. Configured with a 50-ohm Rg resistor, the amplifier achieves a gain of ~400.

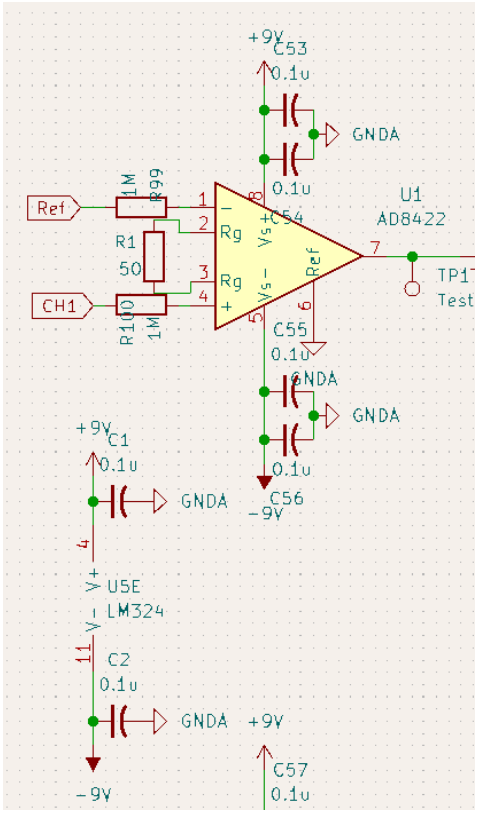


Fig 3. Instrumentation Amplifier for 1ch in KiCAD

Filtering Stages

The signal conditioning chain includes three distinct filtering stages:

High Pass Filter: A 2nd-order Butterworth configuration is used to filter out low-frequency artifacts, set to a cutoff frequency of 0.3Hz.

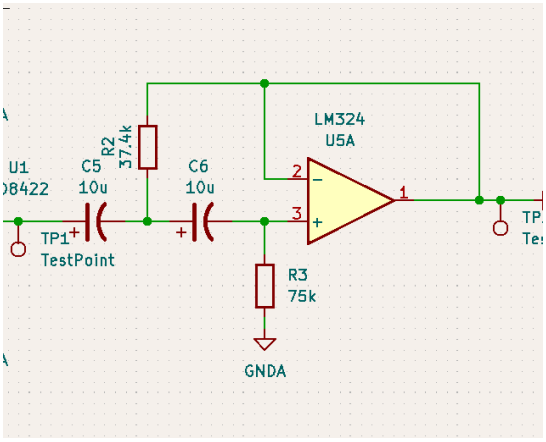


Fig 4. HPF for 1ch in KiCAD

Low Pass Filter: To eliminate EMG signals and high-frequency noise, a Butterworth low pass filter is implemented with a cutoff frequency of 35Hz.

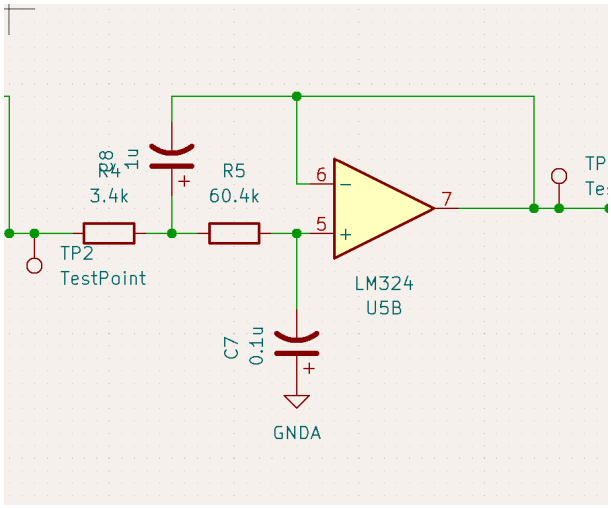


Fig 5. LPF for 1ch in KiCAD

Notch Filter: To mitigate "mains hum" or electric hum associated with alternating current power lines, a filter centered around 60Hz is applied. This stage provides a gain of at least -30dB to effectively suppress this specific noise frequency.

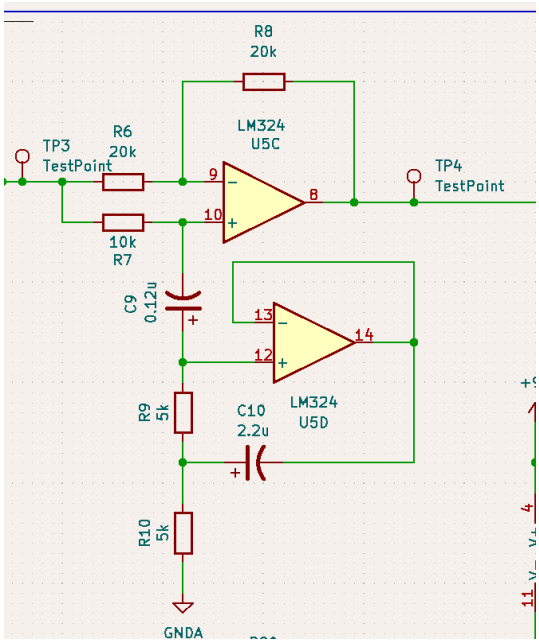


Fig 6. Notch for 1ch in KiCAD

Final Amplification Stage

After filtering, the signal undergoes a final amplification stage using a non-inverting operational amplifier configuration. Through the combination of the initial instrumentation amplifier and this final stage, each channel achieves a total simulated gain of approximately 50,000.

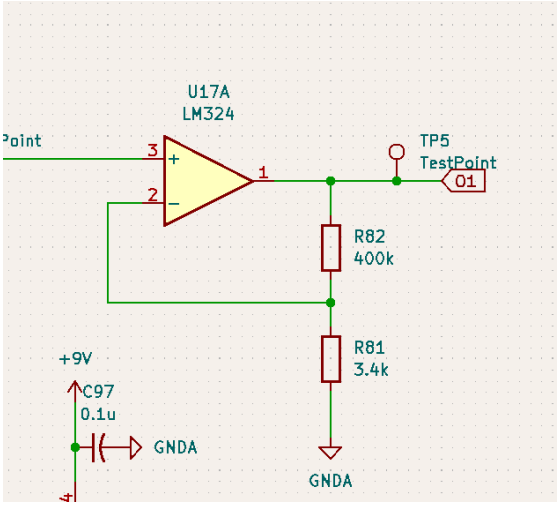


Fig 7. Final Amplification for 1ch for 1ch in KiCAD

Bias and Right Led Driver

High Common-Mode Rejection (CMR) is essential for EEG systems. An RLD circuit was incorporated to set the user's common-mode voltage, thereby increasing the effective CMR ratio. This configuration establishes a grounding standard, preventing voltage loss caused by impedance differences between the subject's reference electrode and the circuit ground.

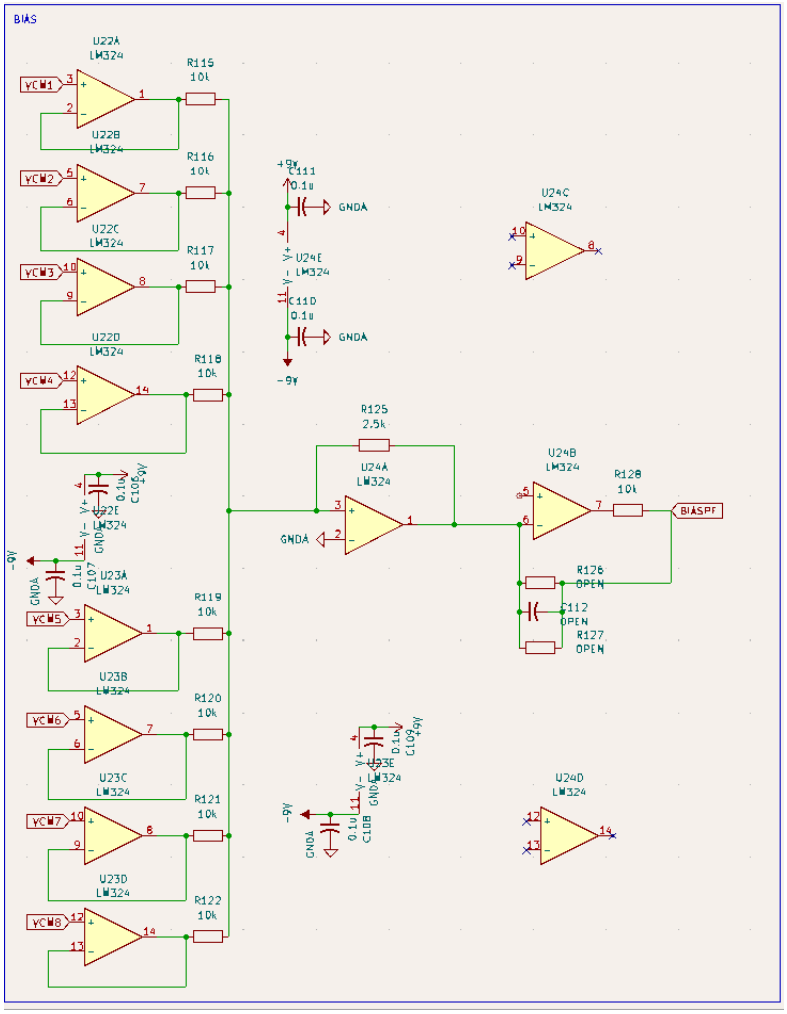


Fig 8. Bias Circuit in KiCAD

Complete Channel

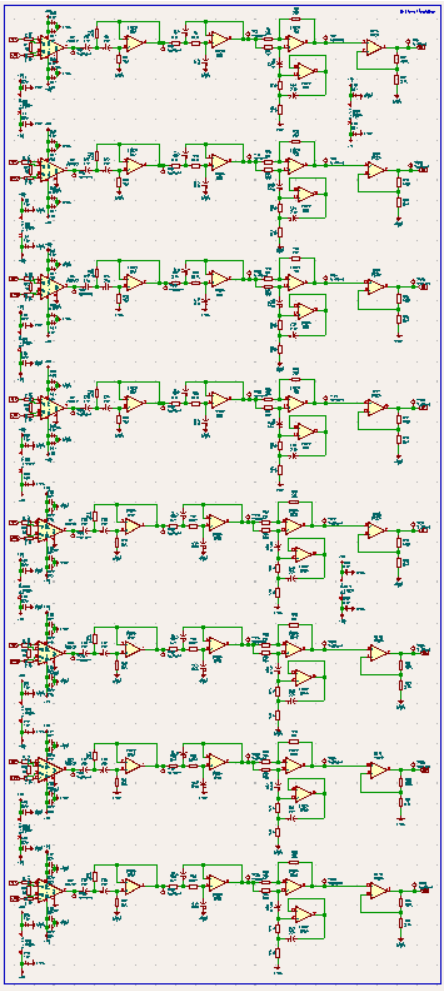


Fig 9. Complete channels (8-channel filtering and amplification)

M1: Power and Digital Front End

Voltage Regulators and References

Power regulation is achieved using a 3.3V Low Dropout (LDO) regulator (LP2985-3.3) sourced from the +9V battery supply, specifically to power the ADC.

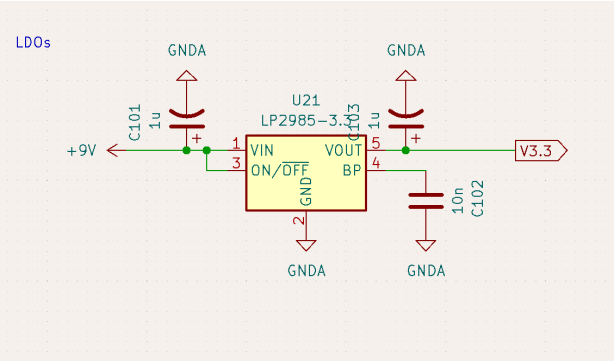


Fig 10. 3.3V Regulator for ADC

Additionally, to ensure the ADC reaches its full dynamic range and maintains precision, a stable reference voltage is provided by the REF3330AIDCKR.

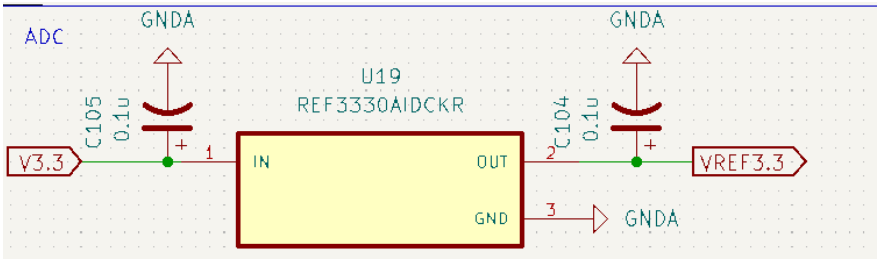


Fig 11. 3.3V Reference voltage

ADC

For analog-to-digital conversion, the system employs the ADS7828, a 12-bit device featuring an 8-channel multiplexer (MUX), a sample-and-hold amplifier, and an internal asynchronous clock. This component utilizes the I2C interface, supporting standard, fast, and high-speed modes.

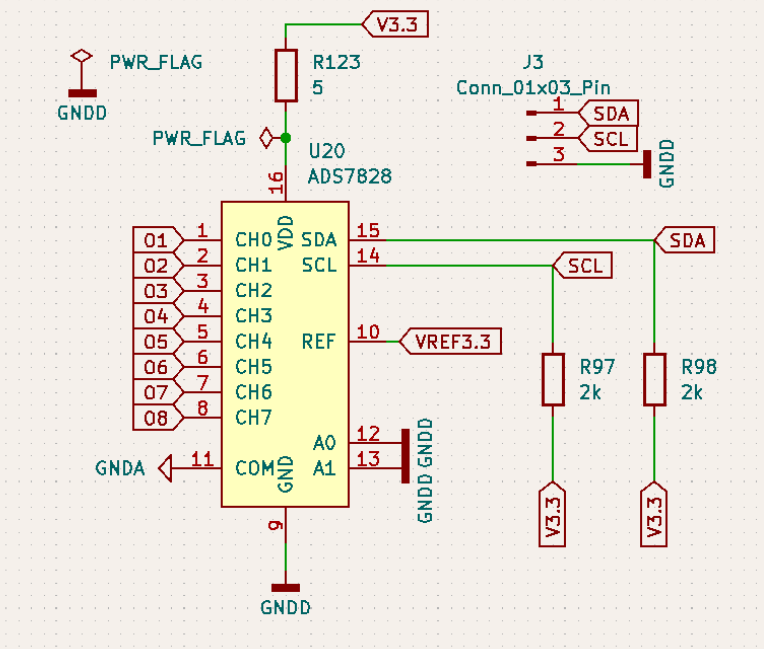


Fig 12. M1 ADC (ADS7828) in KiCAD

Microcontroller Unit (MCU)

In the M1 iteration, the ADC output ideally connects to an external STM32 Nucleo-64 development board. This separation of the Analog Front End (AFE) from the digital processing unit was intentional to minimize digital noise interference that could degrade sensitive analog brain signals. The STM32 Nucleo-64 was selected for its ease of use and resource availability.

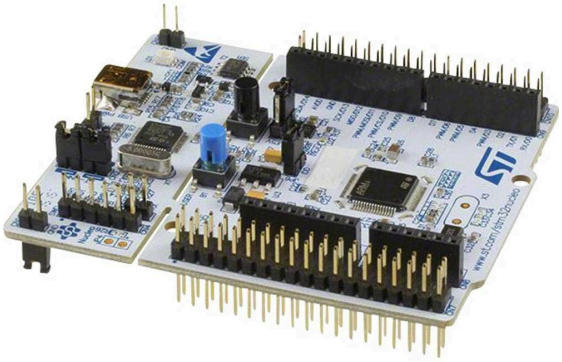


Fig 13. STM32 Nucleo-64 MCU Development Board

Firmware Status

Communication between the ADC and the MCU is established via the I2C protocol (SDA/SCL). Currently, the specific firmware required to process and visualize the data is in the development stage. The logical flow of the firmware will involve initializing the I2C peripheral on the STM32, addressing the ADS7828, and sequentially reading the 12-bit registers for all 8 channels.

M1 Final Schematic and PCB

The schematic and PCB design were executed in KiCad. Due to the high component count necessitated by the 8 discrete channel stages, Surface Mount Devices (SMD) were selected. However, due to the complexity and the high volume of components, the PCB design for the M1 is currently a work in progress and may not be the primary candidate for initial manufacturing compared to the M2.

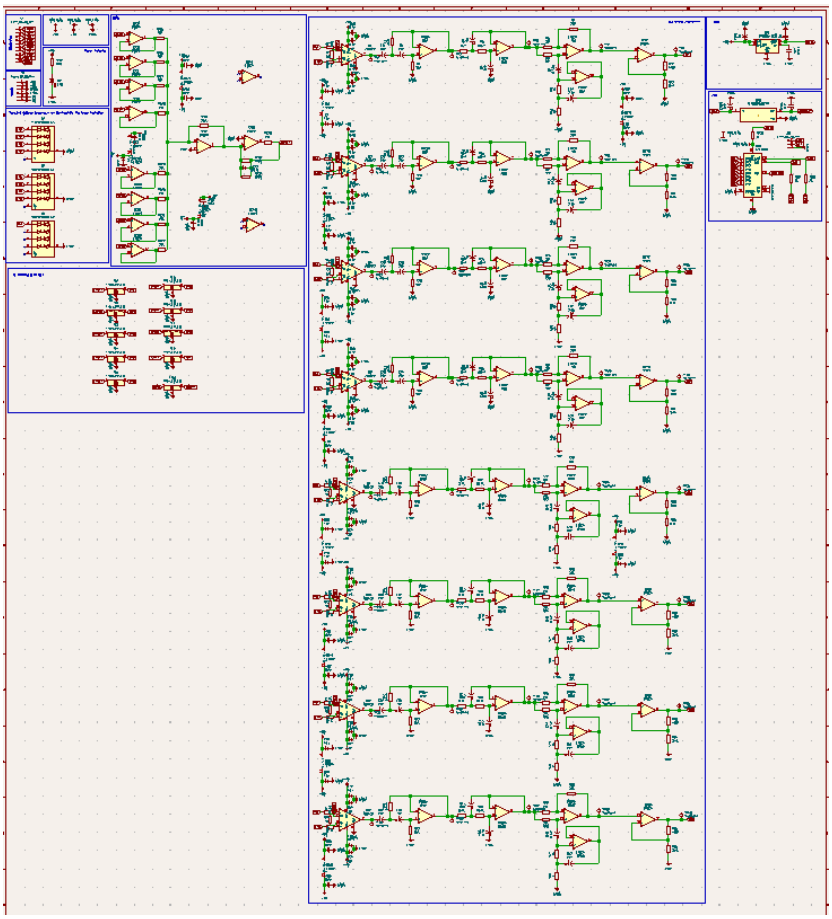


Fig 14. Complete Schematic M1

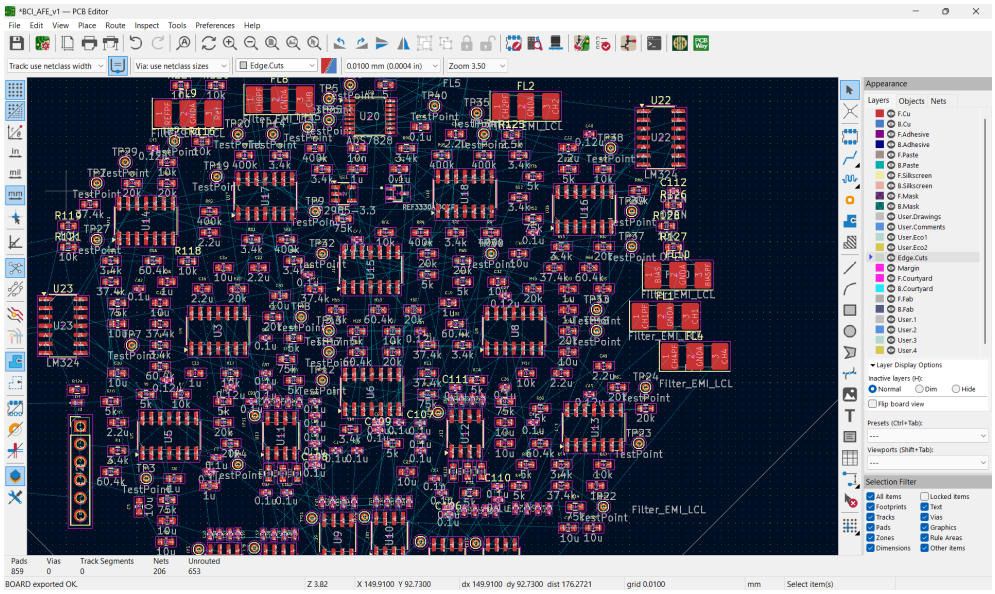


Fig 15. PCB Editor of M1 (In progress)

M2: Integrated Analog Front End (ADS1299)

Overview

Recognizing the scale and complexity of the discrete M1 version, the M2 iteration relies on the ADS1299, a specialized 24-bit Analog-to-Digital Converter designed specifically for EEG and biopotential measurements. This component is widely utilized in systems ranging from hobbyist projects to professional OpenBCI Cyton boards.

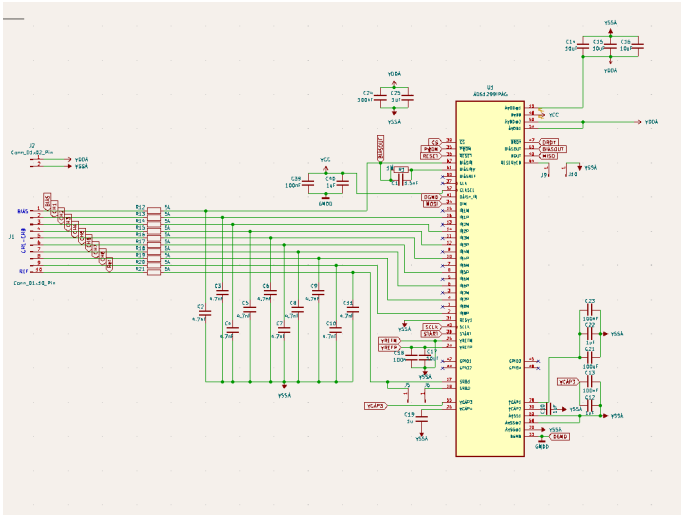


Fig 16. ADS1299 Configuration of M2 in KiCAD

Digital Implementation and Features

Unlike the M1, the M2 incorporates the digital front end directly onto the board. It features an STM32F303CCT Series (Arm Cortex-M4 Core) processor and is powered via USB, eliminating the need for 9V batteries. Additionally, the board integrates a gyroscope to track head movement, providing an additional layer of data for fun EEG experiments.

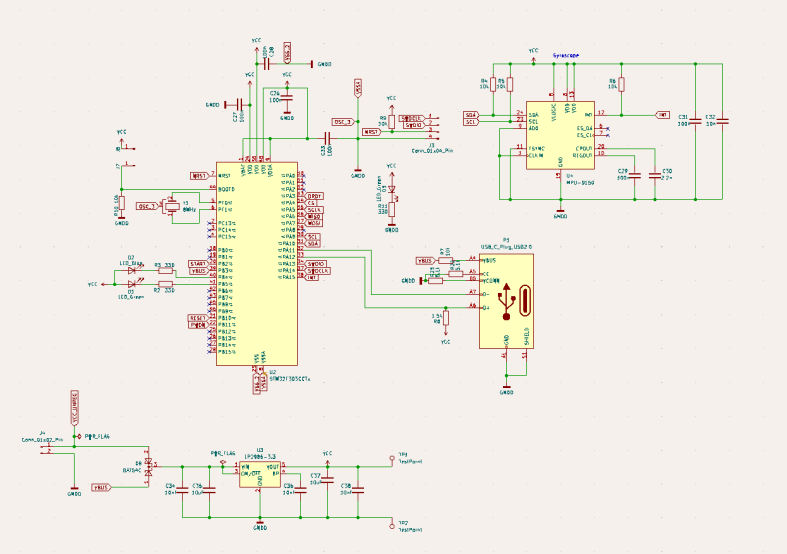


Fig 17. M2 Digital End

M2 Final Schematic and PCB

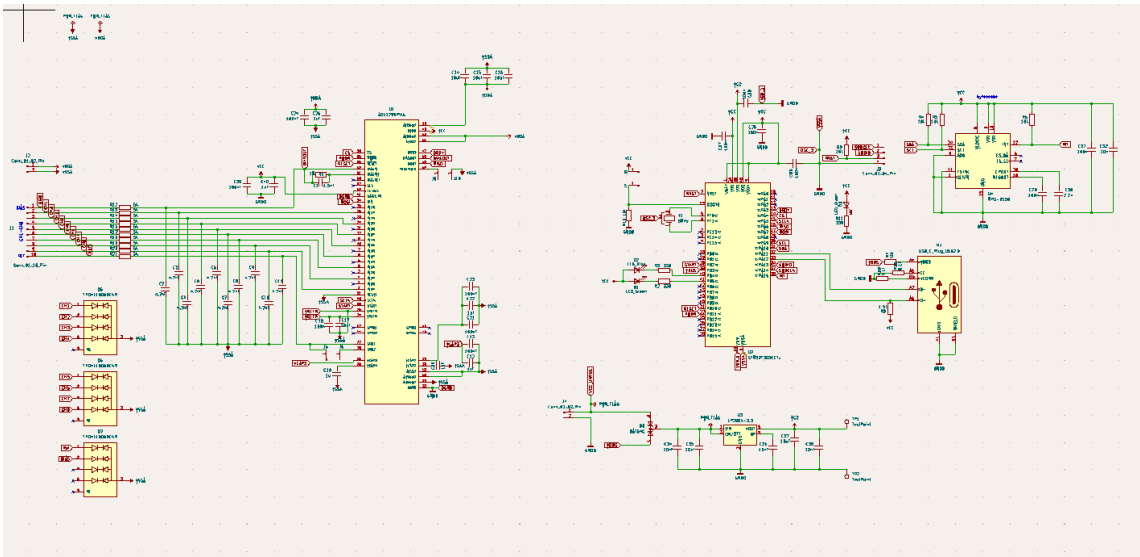


Fig 18. M2 Schematic

Configuration and Implementation

The architecture of the M2 board centers on the ADS1299, a low-noise, 24-bit simultaneous-sampling Delta-Sigma Analog-to-Digital Converter. To achieve high-fidelity EEG acquisition with minimal electrode cabling, the system utilizes a specific "Referential Montage" configuration. This approach leverages the device's internal multiplexing capabilities to optimize signal routing and noise cancellation.

To combat environmental noise (such as 50/60Hz mains hum), the system implements a driven bias, often referred to in biopotential measurements as the Right Leg Drive (RLD). The system does not use BIASIN as a measurement input; rather, it serves as the feedback point for the common-mode signal according to TI, and the internal summing amplifier averages the common-mode interference present on the selected electrodes. This inverted common-mode signal is driven out through the BIASOUT pin to the subject via a protection resistor. This active noise cancellation technique significantly improves the Common-Mode Rejection Ratio (CMRR) compared to passive grounding.

Academic value of project for program

The progress of development of this BCI is not just a personal side project but it could also bring meaningful value to the program. The M1 board, which is designed with strict ground-plane separation and noise management, can provide a good and concrete example of how signal integrity and EMC principles are applied

in real hardware design, more specifically in medical technology applications. This system can also help for instrumentation purposes: achieving micro-volt signal acquisition with discrete components.

Future Work and Conclusion

It was a way harder project than initially thought, it is still not finished as it requires extreme attention to even the tiniest details in order to have a functional BCI, I'll keep working on the project during the winter, have the PCB ready following all mixed signal design and signal integrity standards, refining and minimizing trace impedance and cross talk for signal fidelity and develop the firmware.

Will continue through the winter break with the following objectives:

- PCB Refinement: Finalize the M1 layout adhering to strict mixed-signal design standards, ensuring proper return paths for currents and adequate isolation between the analog front end and the digital switching regulators.
- M2 Fabrication: Proceed with the manufacturing of the M2 board, which offers a more compact and integrated solution via the ADS1299.
- Firmware Maturity: Evolution of the firmware from simple data polling to interrupt-driven acquisition with Direct Memory Access (DMA) to ensure no samples are lost during high-speed transmission.

References

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- http://www.ti.com/lit/an/sbaa188/sbaa188.pdf
- https://www.ti.com/lit/ug/slau443b/slau443b.pdf?ts=1764150777971

Appendix

Files

Ultracortex

Firmware Development (M1 - STM32 & ADS7828)

```
#define CMD_SD_SE             0x80 // Single-Ended Mode
#define CMD_PD_EXT_REF_ON     0x04 // Internal Ref OFF, ADC ON

extern I2C_HandleTypeDef hi2c1;
uint16_t eeg_raw_data[8];

HAL_StatusTypeDef ADS7828_Acquire_All(uint16_t *data_buffer) {
    uint8_t command_byte;
    uint8_t rx_buffer[2];
    HAL_StatusTypeDef status;

    // CH0=0x00, CH1=0x40, CH2=0x10, CH3=0x50, etc.
    const uint8_t channel_map[8] = {
        0x00, 0x40, 0x10, 0x50, 0x20, 0x60, 0x30, 0x70
    };

    for (int i = 0; i < 8; i++) {
        // Construct Command: Single-Ended + Channel + Ext Ref Mode
        command_byte = CMD_SD_SE | channel_map[i] | CMD_PD_EXT_REF_ON;

        status = HAL_I2C_Master_Transmit(&hi2c1, ADS7828_ADDR, &command_byte, 1,
HAL_MAX_DELAY);
        if (status != HAL_OK) return status;

        status = HAL_I2C_Master_Receive(&hi2c1, ADS7828_ADDR, rx_buffer, 2, HAL_MAX_DELAY);
        if (status != HAL_OK) return status;

        // 3. Reconstruct 12-bit value
        // Data is received as: [0000 D11-D8] [D7-D0]
        data_buffer[i] = ((rx_buffer[0] & 0x0F) << 8) | rx_buffer[1];
    }

    return HAL_OK;
}
```